

$$= 3.7 \text{ A}$$

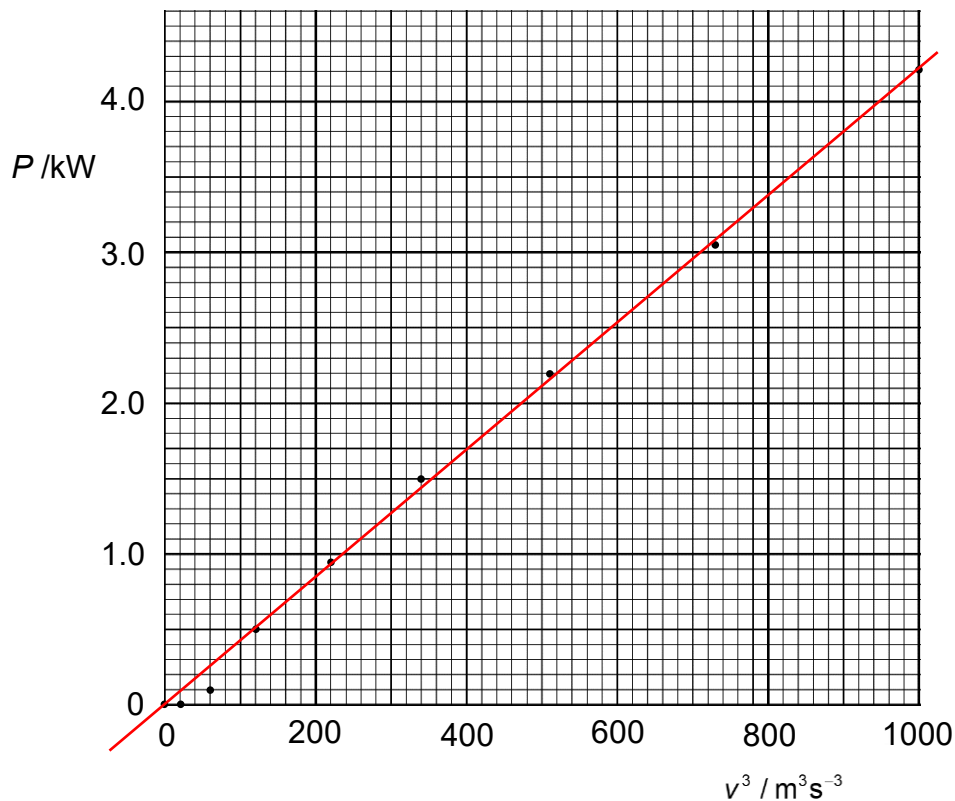
8(a)(i) A, highest output at 10 m s^{-1}

8(a)(ii) the trend for variation of power output with wind speed
still generally increasing

8(a)(iii)
$$\frac{\text{actual power output}}{\text{theoretical power output}} \times 100\% = \frac{4.22}{31.8} \times 100\% = 13.3\%$$

8(b)(i) 1000

8(b)(ii)



point plotted to $\frac{1}{2}$ square accuracy

best fit line

8(b)(iii)

straight line passing through origin

so power output is directly proportional to cube of wind speed

OR

straight line not passing through origin

so power output is linear to cube of wind speed

two points for $v^3 < 100 \text{ m}^3 \text{s}^{-3}$ anomalous because significantly away from best fit line relative to other points

$$8(c)(i) \quad \frac{d_1}{P_1} = \frac{1.0}{0.40} = 2.5$$

$$\frac{d_2}{P_2} = \frac{2.0}{1.6} = 1.25$$

OR

$$\frac{P_1}{d_1} = \frac{0.40}{1.0} = 0.4$$

$$\frac{P_2}{d_2} = \frac{1.6}{2.0} = 0.8$$

percentage difference = 67%

$$8(c)(ii) \quad \ln P = n \ln d + \ln k$$

$$\text{gradient} = n = \frac{y_1 - y_2}{x_1 - x_2}$$

$$= \frac{3.9 - (-0.45)}{2.45 - 0.25}$$

$$= 1.98$$

$$\ln k = \text{y-intercept} = -0.95$$

$$k = e^{-0.95}$$

$$= 0.39$$

8(c)(iii) area swept by rotor blades

$$8(d)(i) \quad A: (2.20)(365 \times 24)(0.028) = \$54$$

$$B: (2.51)(365 \times 24)(0.028) = \$61.60$$

$$C: (2.37)(365 \times 24)(0.028) = \$58.10$$

8(d)(ii) Consider turbine C (with highest cost savings),
In 5 years, cost savings = $(61.6)(5) = \$308$

Since cost savings are lesser than the cost of installation ($\$308 < \3000),
city should not continue with the installation.